

Timestamp 0 Hi everyone, so welcome to this week of the Human-Centered AI class.

Today is the second week where we're going to talk about Human-Centered AI as a design process.

And specifically today we're going to focus on prototyping Human-Centered AI systems.

To give you a bit of an overview, the purpose of this week is to give you the basic resources so you can explore a prototyping method of your choice. So it's going to be quite a short video, unlike last week. Today is going to be quite a short video because when it comes to prototyping, doing is the best kind of thinking.

And the main aim, basically, is to help you understand why we prototype and what are the key principles.

So specifically, we're going to have this lecture is going to be, has three learning goals divided in three parts, more or less. The first one where we're going to understand the basic goal of prototyping, which is about maximizing the rate of learning by minimizing the time to try out new ideas.

We're going to learn some conceptual ideas in prototyping, like the purpose of prototypes, shallow, deep, low fidelity, high fidelity ones, and Wizard of Oz approaches. And then I'm going to present to you some examples of AI-specific prototyping techniques, like Wizard of Oz is one

of those, for example.

so coming back to the last week when we talked about designing human AI systems as a process, I showed you this kind of diagram of the double diamond. We talked last week about discovering requirements and empathizing with our users to defining problems that are meaningful to solve. This week we are on the actual second diamond where we're talking about developing.

Developing and delivering solutions which we're going to keep on testing iteratively.

In the other diagrams that you were introduced to is the phases unsurprisingly called prototyping and testing. We're going to talk about these two, one this week and one next week. Here we're talking about developing and here is the sketching, deciding and prototyping parts of Google's design sprint process.

Okay, but first, you might all have an intuitive idea of what a prototype is or isn't, but I think it might be good to define what it is using the handbook of the HCI. so what are prototypes prototype is one manifestation of a design that allows stakeholders so people are relevant to your design to interact with it and to explore its suitability it is limited that's important in that a prototype will usually emphasize one set of product

characteristics and they emphasize others so prototypes don't do absolutely everything but they emphasize specific aspects of your system that you actually want to test with users, for instance, or you want to sketch them out by yourself to get feedback from you or from others. And with this definition, prototypes can be anything, as you can imagine. They can be a paper-based storyboard all the way through a complex system software. It can be a cardboard mock-up or a pressed piece of metal if we're talking about physical things. And the purpose here of prototypes is to help users envision the important things of interactions with these newly designed systems. So again, these are early models of interaction. They're not full implementations. They focus experience on experience over the algorithm, for instance, in the early stages. And they're useful for iteration, validation, as well as user testing that we're going to discuss as well. And they're very closely related to how you can Timestamp 4 minutes include users in your design process. So the things we were talking about last week, very often we use prototypes as probes, for instance, to give them to users to give you feedback on your idea, not necessarily

to give you feedback on very minuscule parts of the idea, but on the whole kind of problem that the users might be facing.

So why prototype? Prototypes can help you think through and understand the possibilities of your design space

and they help you attest assumptions about your user, their experience or even if the technologies that

you're considering are actually viable. So this is actually quite crucial here.

first because they're about yourself so when you start prototyping you can actually test your own ideas and think through what you have

been discussing or thinking about doing and the second part is you can also they're very social sociable in

the sense that you can use them to to ask other users or other people around you feedback on the ideas that you have might have only had in your mind until then.

Now specifically when we're talking about human-centered AI,

why would you choose to prototype? So you might want to make sure that you want to be able to align the system behavior with the user expectations, which is quite obvious and it's

also something that we have discussed last week. You might want to catch usability issues early, so you make kind of mismatches and mental models you might have discussed that might be something you might be easily to find through

prototypes and you might also want to explore ethical trust and explainability concerns before implementation and that crucially comes back to this type of prototyping where it would be more critical design or speculative prototyping which we can discuss a little bit as well so when we talk about traditional AI if you want to kind of compare this like how you've been Timestamp 6 minutes doing things potentially until now and how will you make your process more user centric centered like your traditional ai might be model centric whereas your human centered would be human centric a bit obvious in the traditional ai you might be putting your data first like you have a large data so you might be working from the assumption that it's a useful one whereas in human centered ai you would want to see what are the needs of the people that and then you would go find the data for that Traditionally, AI would be evaluated by accuracy of your model, for instance, whereas in human-centered AI, you might evaluate by usability, how usable the results are, the models are, the results, how trustworthy they are, or how fair, and other topics like this, which we will discuss in the upcoming weeks as well. now to come a little bit into some uh terminology to help us all communicate in the same way they're uh more

or less like prototypes are usually divided into two types of prototypes these are the most common kind of division you would talk about when you talk about prototyping which would be low and high fidelity ones so low fidelity prototypes are they have some advantages and disadvantages, both of them, right? So low-fidelity prototypes would have lower development costs, would be easier to evaluate multiple concepts very quickly.

They're less constrained, so they're more conceptual rather than actually the nitty-gritty of the implementation. And they explore high-level design layout options.

But at the same time, there's not really a possibility for error checking. There is little detail. They cannot necessarily test dynamic layout although that's changing with some of the tools i'm going to introduce to you where and these are things like i'm going to discuss give you some examples of low fidelity prototypes but these are things like paper prototypes or mock-ups are low fidelity prototypes

Timestamp 8 minutes whereas high fidelity prototypes are at least partially functional they're often interactive they have like a content like effective preview of the layout you the look and feel resembles

that of the final product,
even though it might
not be the actual final
one you might be giving.

And it's often used
as a marketing or
sales tool where
you want to kind
of introduce it to
potential clients.

The disadvantages of high
fidelity prototypes, however,
is that, as you can imagine,
it takes kind of time
and resource intensive to
develop them. It's much
faster to sketch out some
boxes rather than to design
a proper layout. They're
not really effective for
requirements gathering in the
early stage of ideation and
they may confuse stakeholders
or your users, potential
users that you're testing
them with or you're
showing them to, that this
is how your final prototype
would look like, what the
final product would look like,
rather than allowing them
to imagine it as a prototype
that they can actually
engage and critique with.

so looking a little bit on
the level of fidelity as
we were talking about you
might have sketches ideas
and this is a really interesting
diagram right because
it shows you if you see
here you have on the on
the bottom you have
different types of prototypes
and what they communicate
so sketches wireframes a
prototype as you an interactive
one as you would say
here or the final design
and then on the top on the y

-axis you have the commitment and level of fidelity
so when you make a sketch you it has kind of low commitment in the sense that you don't mind to it took you five minutes to make or three minutes to draw so it's very easy to just throw it away and make a new one or make a series of sketches to be able to kind of explain the possibilities that you might have so this is a good way to communicate ideas or possibilities that you are able to pursue because it has kind of low commitment and you can iterate on that quite quite easily you can make wireframes as the next stage where this would be good for concept validation where let's say you have some idea of an app you want to create with a AI messaging function, for instance, then you can easily create the structure and the overall flow of how your app might work. And that would allow you to also do quite quick iterations. So it's also, again, quite low commitment in the sense that you can draw something like that quite quickly and then show it to people to get feedback on the concept. When you start refining that, you might make a high fidelity prototype this is what the prototype kind of refers to here the word prototype refers to here a high fidelity one where you would have like the

structure like you have in the wire frame you would have the flow and you would have some details even if the the details are not like the final visual design but it might even have some interactivity for instance like if you click a button then something actually might a new page might appear or something like this and that would be for refinement, also testing some more complicated features that are hard to test kind of individually. And finally, like you have your final design, your fully branded and fully developed product that you can, as we will learn next week, you can actually do some usability testing to it, where this is the final actual kind of iteration where you have everything fully developed and then you can actually give it to people to get feedback upon and do a b testing or do any kind of kind of more wide scale testing or evaluation that's the word i'm looking for for wide scale evaluation of your design the point here of this graph is that if you want to just ideate or you want to test what kind of ideas

Timestamp 12 minutes to use the point is to make sketches are actually up to wireframes you're not going to make a fully -fledged interactive prototype just to test an idea.

So you have to think about strategically what

point in time are you going to use what kind of prototyping or sketching or wireframe technique.

Okay, so let's go into this within a bit more detail now.

Let's talk about low-fidelity prototypes.

Okay, so one type of low-fidelity prototypes are wireframes. And wireframing is a process where designers basically draw overviews of interactive products to see the structure and the flow of possible design solutions so here you see an example of that in myro in a myro board where the big kind of square with the x implies that there is an image there and then there is some text and there's a headline and then it kind of tests out like draws out sketches out different layouts for this i guess this is a normal website kind of thing and then And you can give them to other people to actually test to see what makes sense to use, to get feedback from.

So wireframes are primarily tools for collaboration towards making better prototypes and products faster. So before you start drawing the actual content you're going to use, you're probably going to start doing something like this.

And there are tools that you can use to do these wireframes. this is one it's quite already it's called balsamic where you have these kind of little boxes and

elements that you can just kind of drag around and make your your your page like or your app so another typical way that you can prototype things low fidelity and cheap and quite fast is paper prototyping as you can see here you mostly have potentially wire frames or even things that you have printed out um to like text you have printed out and then you actually you sketch out potential concepts or flows or ideas on paper but then the important part here is that you actually test those with your potential users so your user you give this to your potential user and then you ask them to give you feedback about how does that seem to them in research it's also quite typical to to see these kind of fake cardboard ipads or tablets or or phones where you would kind of insert these paper prototypes and you would allow user to to play with them in a way and it's one might ask like why would you do this in paper because it comes closer to the experience of what the the participant might be doing so if you have it in a in a way that they are familiar with like the way that they might be end up using it in reality also then it's more chance that the feedback that you're going to get is more authentic and also because it looks so low stakes like it's just hand drawn with pens or

printed out quickly with cheap paper then research has shown that there's more chance that your users you're you're testing with that they feel like they have more agency to change or to comment or to critique these ideas because they're obviously not fully fledged in this project and nowadays it's kind of more common to do these kinds of prototypes digitally just because it takes it would take the similarly amount of time to similar amount of time to to draw and cut out and create these paper paper prototypes than to create three four frames in Miro and put some blocks there that connect we also have for for these kind of low-stake prototyping you can use Miro for wireframes but we're also going to be talking about Figma for for having some fast interactions so what you also have are click-through prototypes as I mentioned before even though paper prototyping can be very powerful nowadays there are also other ways to do quite fast prototypes that are easy low stakes prototypes that you can test with users so like click through prototypes with platforms like figma or azure or balsamic as we talked about but we're going to talk about figma in specific so you see here what a quite easy it doesn't require really much programming It's just you connect these little arrows to different frames and you can actually make a prototype.

So for instance,
you can have boxes,
like you create boxes
like this and with
a simulating button.
So you can see it's
very, very simple.
It's very low fidelity.
And by connecting it to
and clicking through the
interaction details in the
prototype phase, then you can
quickly make a prototype
like what you see on the
on the right where you
press this button that this
button and it comes on the
screen becomes all blue and
you press the red button
and the screen becomes all
red so the the point
here being i just want to
demonstrate that uh figma is
quite fast and figma and other
tools nowadays are quite
fast and quite easy to to
generate interactive prototypes
that you can literally
test on your mobile on
your screen so also on your
mobile devices because
they're apps that you can
actually simulate that and
you can evaluate your ideas
quite quickly and you can
just draw very low level
boxes like wireframe style
to to express your ideas
so let's talk a little
bit about high fidelity
prototypes so as we said
high fidelity prototypes are
something you would do
later on you would probably
start with something low
fidelity you would prototype
your ideas you would try
to test your assumptions
quite quickly and you would
find specific assumptions

to test you would make something you would show it you You would use maybe Wizard of Oz or you would use like click through prototypes and then once you have a clearer idea of what you want to create, then you will create more high fidelity prototypes and then there are two types of high fidelity prototypes. You can do vertical prototyping or horizontal prototyping and that's a little bit about the approach you're taking. So with vertical prototyping, you are basically focusing whatever is here as a plus is fully implemented and whatever is X is not implemented. So you see the visual language here. So we're talking about the vertical prototyping where only a few unselected functions are actually being implemented so that you can see quite realistic tasks. So, for instance, if you are testing a new messaging app, perhaps the messaging function works perfectly, but everything else, messaging between friends works perfectly, but everything else is kind of not really implemented because it's beyond the point. You want to test the specific function of that, of that specific feature. horizontal prototyping on the other hand is when you are all the visual parts of the user interface of an application or a computer system are actually implemented

but they're all they're quite partially implemented they they might be shallow like they might no data might be processed for instance so in the in the example of the of the messaging app perhaps all the interface you see you can add friends you can you can click through all the different buttons you see on your high fidelity screen but nothing gets saved or no real message gets sent at that moment so you're you're basically testing different things in this case so coming back to click-through prototypes like figma is something that you will use quite a lot for your high fidelity prototypes So we suggest that you start with Miro to do your wireframes and then perhaps do small prototypes, interactive ones in Figma to test some quick assumptions. And then in the end you're going to have a more complex Figma file with, as you can see, it can become quite intense in terms of interactions and buttons and all the possibilities that you can have.

Finally, something we haven't touched upon but is actually also quite fundamental in human-computer interaction is physical computing.

So there are quite a few platforms out there like Arduinos, Raspberry Pis, 3D printing, laser cutting. there are quite a lot of techniques now for quickly creating form, prototypes that need,

that are three-dimensional,
or that are interactive
in another way, with
electronics for instance,
and this is a whole kind
of research domain in
human-computer interaction,
that includes also
different types of
expertise than the digital
applications that we are
working on right now.

So here you see some examples
on the top left of an

Arduino that is actually
being prototyped to make a
stopwatch. I think in this
case it was prototyping a game.

on the bottom you see
how exoskeleton might be
prototyped we're using 3d
printing on the on the laser
you see with the hexagons
is actually a laser

cutter that you can really
quickly cut wood or vinyl
or not vinyl sorry wood
or acrylic or other types
of materials that you
can make quick and dirty
prototypes like the ones
you see on top which is
actually an interactive
lamp and physical computing
also allows for for things
like tangible interfaces
and and like for museums or
for other settings like that
another very um useful
tool for for low fidelity
prototyping is this is kind
of related to low fidelity
prototyping because it has
to do with the evaluation of
prototypes is wizard of oz
prototyping and the wizard
of oz prototyping refers
to the the wizard of oz uh
film where for those of
you who who might have not

seen it there is this um
seemingly powerful wizard that
knows everything and
commands everyone whereas in
Timestamp 24 minutes fact they find out that
it's just a person behind a
behind a curtain kind of
manipulating the scenes and
making it seem like it's a
wizard a powerful wizard so
they take this analogy of
the metaphor of you you make
something seem as it's
more complicated than it
actually is in reality and
we use that in technology so
this was the user wizard
of testing for instance in
ibm when they were testing
out the listening typewriter
which is really interesting
if you think about where
we are right now in terms
of technology this is from
1984 so they wanted to
understand the experience of
being able to do speech to
text systems so they had
a person in one room and
they had a screen and they
were given a screen and a
microphone and they were told
that if you speak into that
microphone then that the
computer will magically
or anyway would kind of
transcribe the speech into
text and you would see that on
your computer so they wanted
to test that experience
if it would make sense to
do and but in reality they
didn't have the technology
to be able to do that
so the wizard of oz did
the wizard in this case was
another person behind the
scenes in another room
that was actually typing
out what they were hearing

from the um what they were
hearing from the microphone
so the point here is
that this is much lower
stakes this is low fidelity
prototyping you can very
quickly test out ideas
that you might either not
have the technology for at
the moment or the technology
might be to might take
some effort or some time
so much effort to make
that you want to test the
experience of it before you
actually implement it So
Timestamp 26 minutes a lot of this literature
on prototyping comes
from human-computer
interaction, but in this
case, in this lecture,
we want to look at how we
can prototype specifically
AI kind of systems.

So we can use the
same methods, and I
have some examples
here, for instance.

So if you have like an AI
use case of a recommender
system, then you can actually
make a mock interface
with Figma or even with
paper if you want to,
with fake suggestions
and then give
that to the participants
and your users
and kind of see
how they respond.

If you want to have a use
case, if you have a use case
of AI where they were
talking about a chatbot or
some kind of LLM, for
instance, then something before
you even try fine tuning or
training it, then you can
actually do a Wizard of Oz
with scripted responses and

see how they respond how
people respond to it and
that would be especially the
case for like niche domains
like for instance for a
nursing AI app or a chatbot
or some kind of agent in
a hotel for instance and
then it will make more sense
to do it first with the
robots than actually building
the model and then another
use case might be having
a classifier UI where here
you might prototype it by
adding kind of confidence
bars or explanation cards
to actually give like some
background information
to participants about
how how they how the
system works for instance
so there are two main things
to remember when we're
thinking about how to decide
the prototype like each
prototype you create you
have to remember what
our assumptions are your
prototype testing and what
does this mean for how and
why you prototype so you're
Timestamp 28 minutes not gonna prototype in
the same way every single
system you're gonna choose
sometimes low fidelity
sometimes deep prototype
sometimes shallow prototypes
and you're gonna kind of
decide which ones are more
relevant to your context
depending on the assumptions
that you're trying to
test so with a chatbot
you might want to test the
interaction or the fact
that do you want to personalize
it or do you want to
have like a human-like
agent talking and then you

would test it in that
way for instance you will
prototype it with different
types of characters so yeah
remember that when you're
working on your projects